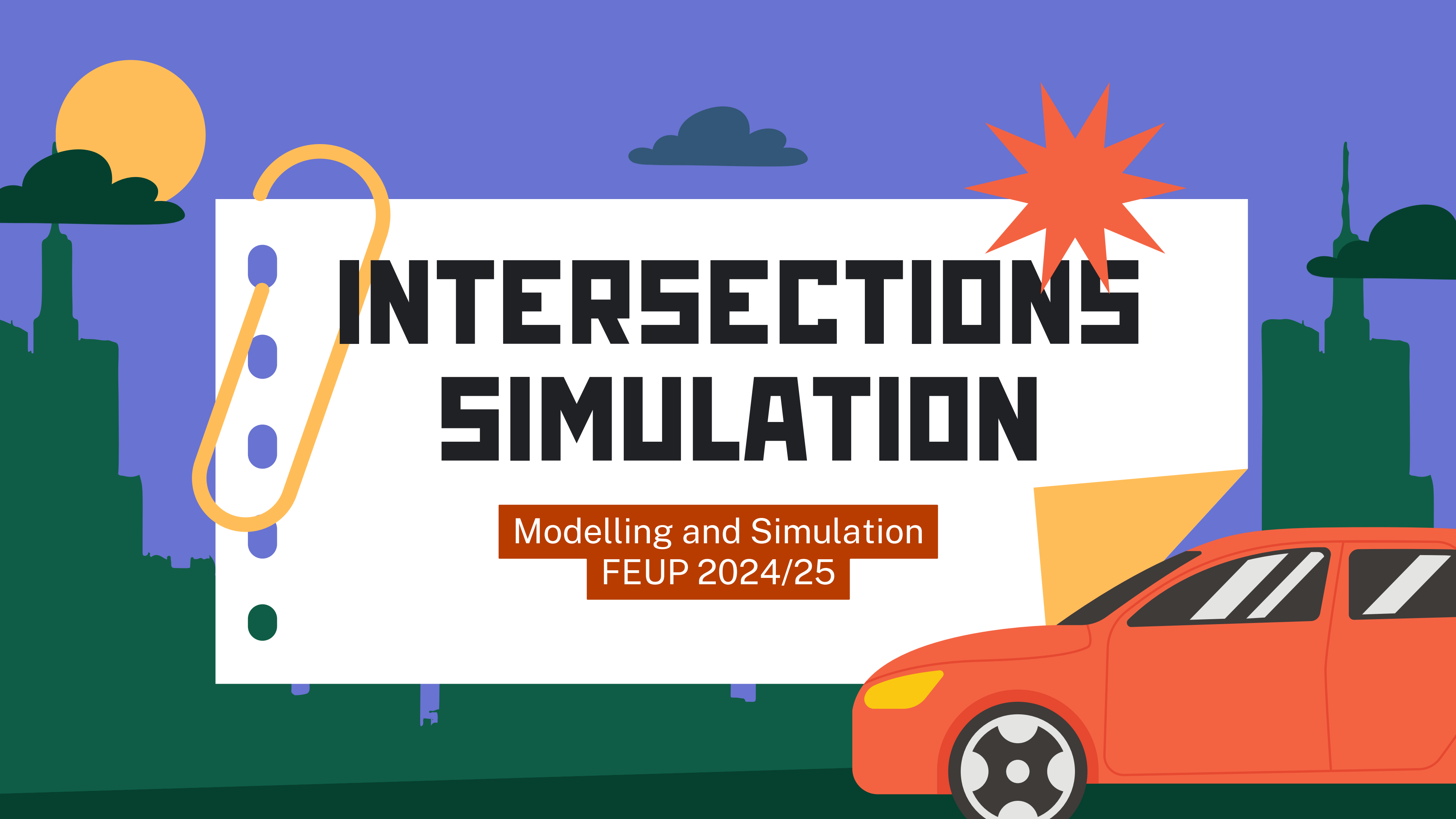
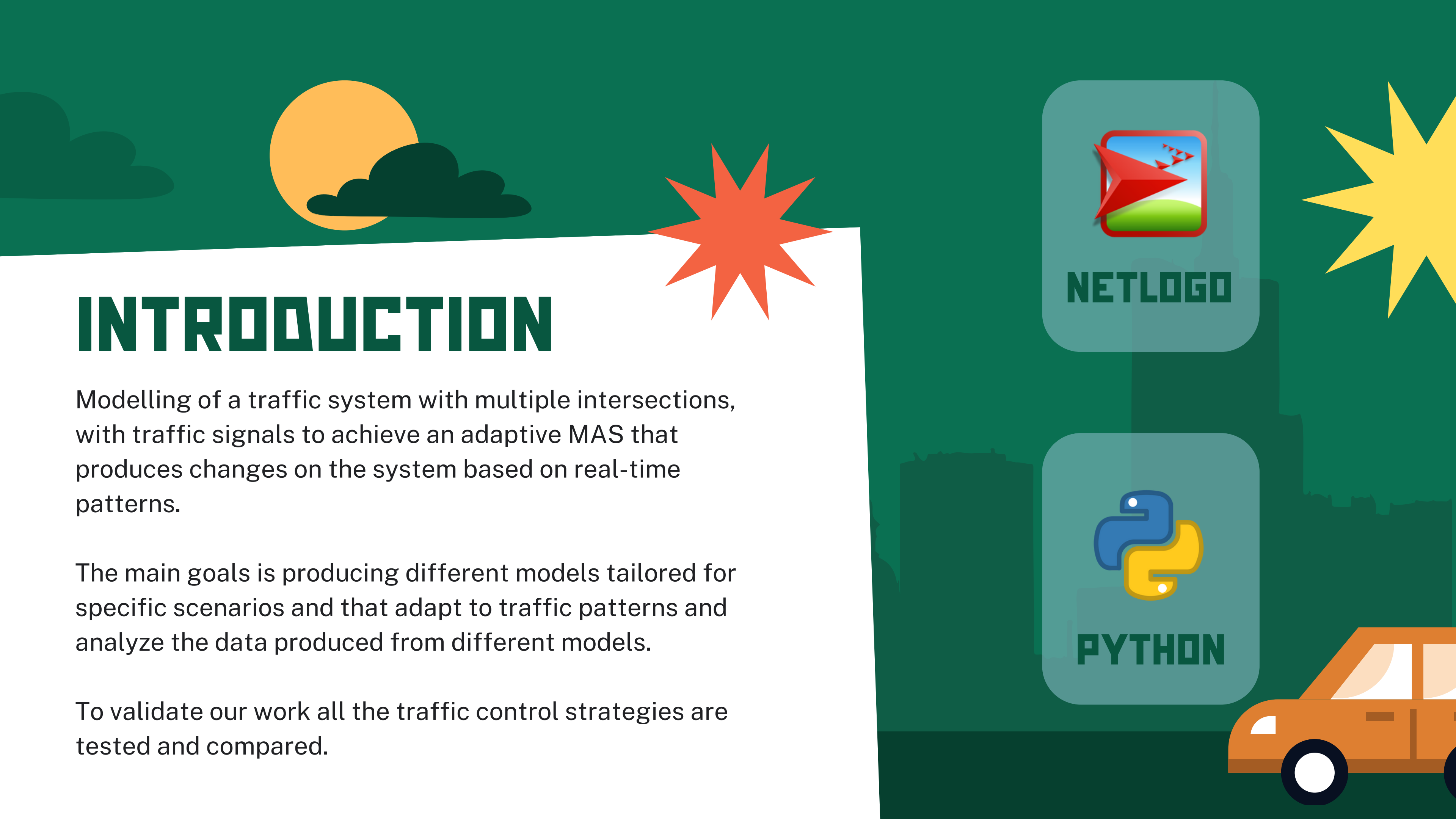




INTERSECTIONS SIMULATION

Modelling and Simulation
FEUP 2024/25



INTRODUCTION

Modelling of a traffic system with multiple intersections, with traffic signals to achieve an adaptive MAS that produces changes on the system based on real-time patterns.

The main goal is producing different models tailored for specific scenarios and that adapt to traffic patterns and analyze the data produced from different models.

To validate our work all the traffic control strategies are tested and compared.



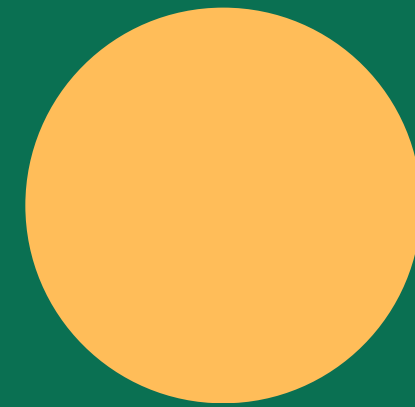
NETLOGO



PYTHON



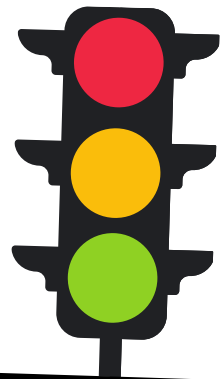
ENTITIES



SYSTEM VARIABLES



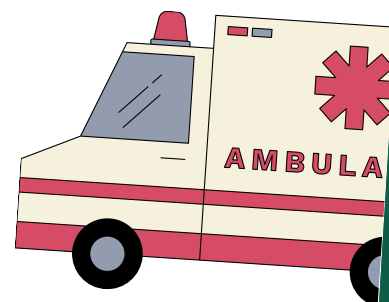
Vehicles



Traffic Signals



Intersections



Priority Vehicles

Traffic Flow

Waiting Time

Signal Timings

Traffic Density

Priority Score

VARIABLE 1

DENSITY

Roads with more vehicles on them have priority to have a green light, avoiding many vehicles from being stopped

In this way, if a road is empty it should not have green lights, if the other road on the same intersection has vehicles on it



VARIABLE 2

WAITING TIME

Roads with higher waiting time from the vehicles on them have priority to have a green light, to ensure fairness.

In this way, if a road with a single car hasn't had a green light in a few time the light should turn green for that road, even if the other road in the intersection is full.



VARIABLE 3

PRIORITY VEHICLES



When a priority vehicle is on a certain road, the traffic light should turn **green** as soon as possible and the other traffic light on the intersection should turn **red**

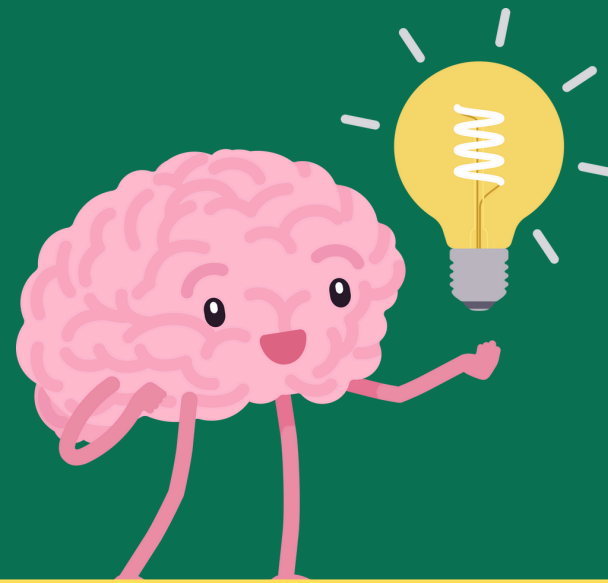


If there are **multiple priority vehicles on the same intersection**, the **first** one to **arrive** will have priority first



Priority vehicles are defined by their color: **all red vehicles have priority** and there is no vehicle that is not red that has priority

VARIABLE 4



PRIORITY SCORE

$$RPS_R = W_1 \cdot D_R + W_2 \cdot RT + W_3 \cdot RP$$

WHERE:

RPS -> Road Priority Score

R -> Road

D -> Density

T -> Waiting Time

P -> Amount of Priority Vehicles

W -> Weight of Variable

VARIABLE 5

COMMUNICATION BETWEEN INTERSECTIONS

GLOBAL PRIORITY SCORE (GPS)

$$GPS_R = \sum_{R \in I} RPS_R \cdot 1 / q$$

WHERE:

R -> Road of same direction

RPS -> Road priority score

D -> Distance

I -> Intersection

OPERATION POLICIES TESTED

CASE 1

FIXED-TIME CONTROL

Traffic signals operate on a predetermined schedule without adapting to current traffic conditions.



CASE 2

ADAPTIVE TRAFFIC SIGNAL CONTROL

Traffic signals adjust their timings based on real-time traffic data from sensors. It can be density, the car's waiting time or a combination of these factors.

CASE 3

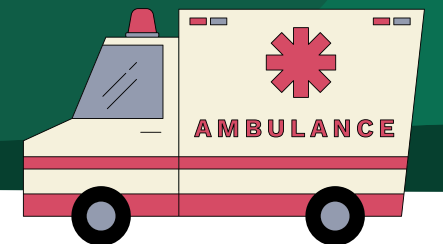
MULTI-AGENT SYSTEM

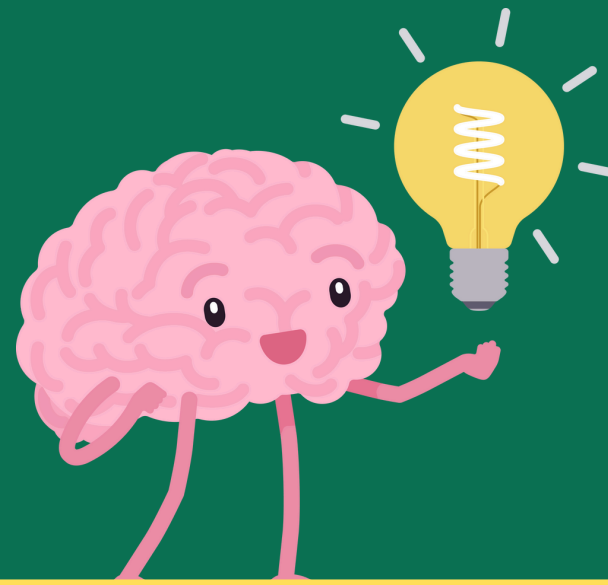
Traffic signals act as autonomous agents that use decision-making algorithms to optimize signal timings, considering information from neighboring intersections.

CASE 4

PRIORITY-BASED CONTROL

Priority vehicles receive prioritized green lights, potentially at the expense of increased waiting time for other vehicles.





SOME RESULTS

INPUT/OUTPUT

CASE 1

Strategy	Green Length	Yellow Length	Neighbors Asked	Weights	Total Traffic Input	Total Traffic Output	Total Waiting Time	Average Total Waiting Time	Total Accidents	Priority Average Waiting Time
auto?	12	1	-	-	10175	10159	90804	8.92	0	9.38
auto?	1	1	-	-	10162	10139	124416	12.24	0	11.68
auto?	1	5	-	-	7632	7553	698479	91.52	0	94.41
auto?	2	5	-	-	8508	8420	711610	83.64	0	82.99

CASE 2

Strategy	Green Length	Yellow Length	Weights	Total Traffic Input	Total Traffic Output	Total Waiting Time	Average Total Waiting Time	Total Accidents	Priority Vehicles Count	Priority Average Waiting Time
priority-score?	-	2	(50, 40, 10)	10224	10178	29530	2.89	15	200	2.58
priority-score?	-	1	(10, 50, 40)	10209	10130	11882	1.16	35	207	0.73
priority-score?	-	5	(30, 40, 30)	2721	2575	1377283	506.17	0	60	351.5
priority-score?	-	5	(25, 25, 50)	2786	2639	1359023	487.8	0	48	595.88

INPUT/OUTPUT

CASE 3

Strategy	Green Length	Yellow Length	Neighbors Asked	Weights	Total Traffic Input	Total Traffic Output	Total Waiting Time	Average Total Waiting Time	Total Accidents	Priority Vehicles Count	Priority Average Waiting Time
multi-agent?	-	3	1	(60, 0, 40)	10159	10125	59655	5.87	8	209	5.08
multi-agent?	-	1	2	(80, 20, 0)	10174	10099	11877	1.17	33	194	1.26
multi-agent?	-	1	2	(0, 0, 100)	1751	1670	802430	458.27	0	36	0.33
multi-agent?	-	2	2	(0, 0, 100)	1751	1670	802430	458.27	0	36	0.33

CASE 4

Strategy	Green Length	Yellow Length	Neighbors Asked	Weights	Total Traffic Input	Total Traffic Output	Total Waiting Time	Average Total Waiting Time	Total Accidents	Priority Vehicles Count	Priority Average Waiting Time
allow-priority?	3	2	-	-	10163	10136	137953	13.57	0	203	1.65
allow-priority?	9	3	-	-	10151	10110	185352	18.26	3	178	3.17
allow-priority?	4	5	-	-	108	27	804883	7452.62	0	3	3313.67
allow-priority?	2	2	-	-	109	28	804817	7383.64	0	4	2481.25

AVG. WAITING TIME

CASE 1

Strategy	Green Length	Yellow Length	Neighbors Asked	Weights	Total Traffic Input	Total Traffic Output	Total Waiting Time	Average Total Waiting Time	Total Accidents	Priority Vehicles Count	Priority Average Waiting Time
auto?	11	1	-	-	9944	9932	87374	8.79	0	-	8.64
auto?	9	1	-	-	9968	9948	88347	8.86	0	-	8.62
auto?	1	5	-	-	7632	7553	698479	91.52	0	-	94.41
auto?	2	5	-	-	8508	8420	711610	83.64	0	-	82.99

CASE 2

Strategy	Green Length	Yellow Length	Neighbors Asked	Weights	Total Traffic Input	Total Traffic Output	Total Waiting Time	Average Total Waiting Time	Total Accidents	Priority Vehicles Count	Priority Average Waiting Time
priority-score?	-	1	-	(50, 0, 50)	9989	9972	7752	0.78	6	244	0.19
priority-score?	-	1	-	(40, 0, 60)	9943	9924	7959	0.80	5	209	0.19
priority-score?	-	5	-	(50, 20, 30)	2990	2823	1554873	520.02	0	56	344.29
priority-score?	-	5	-	(33, 34, 33)	2884	2721	1475482	511.61	1	57	355.82

AVG. WAITING TIME

CASE 3

Strategy	Green Length	Yellow Length	Neighbors Asked	Weights	Total Traffic Input	Total Traffic Output	Total Waiting Time	Average Total Waiting Time	Total Accidents	Priority Vehicles Count	Priority Average Waiting Time
multi-agent?	-	1	1	(20, 0, 80)	9913	9879	10195	1.03	12	187	0.15
multi-agent?	-	1	1	(40, 10, 50)	9982	9931	10303	1.03	22	177	0.39
multi-agent?	-	5	2	(40, 40, 20)	2983	2815	1518519	509.06	0	55	183.42
multi-agent?	-	1	2	(0, 0, 100)	1751	1670	802430	458.27	0	36	0.33

CASE 4

Strategy	Green Length	Yellow Length	Neighbors Asked	Weights	Total Traffic Input	Total Traffic Output	Total Waiting Time	Average Total Waiting Time	Total Accidents	Priority Vehicles Count	Priority Average Waiting Time
allow-priority?	3	1	-	-	10012	9999	89308	8.92	0	206	1.11
allow-priority?	4	1	-	-	10093	10072	94408	9.35	0	203	1.10
allow-priority?	4	5	-	-	108	27	804883	7452.62	0	3	3313.67
allow-priority?	2	2	-	-	109	28	804817	7383.64	0	4	2481.25

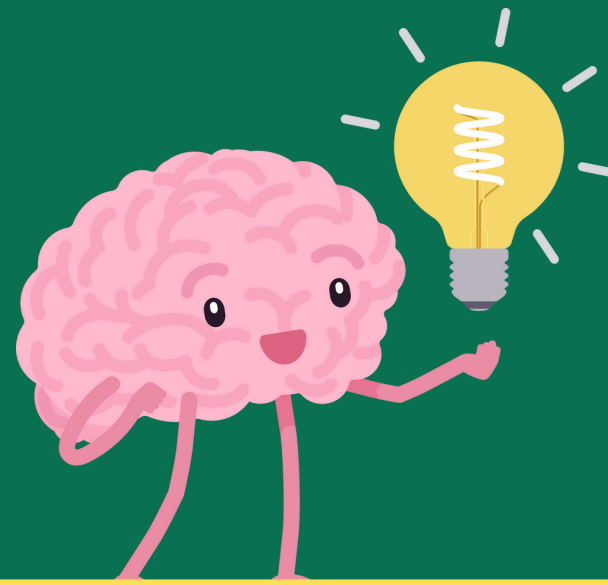
AVG. PRIORITY VEHICLE WAITING TIME

CASE 3

Strategy	Green Length	Yellow Length	Neighbors Asked	Weights	Total Traffic Input	Total Traffic Output	Total Waiting Time	Average Total Waiting Time	Total Accidents	Priority Vehicles Count	Priority Average Waiting Time
multi-agent?	-	1	1	(0, 0, 100)	5058	5029	269246	53.23	0	98	0
multi-agent?	-	2	1	(0, 0, 100)	5058	5029	269246	53.23	0	98	0
multi-agent?	-	5	2	(40, 50, 10)	3631	3468	1331307	366.65	0	76	684.67
multi-agent?	-	5	2	(20, 80, 0)	3432	3272	1422096	414.36	0	81	669.98

CASE 4

Strategy	Green Length	Yellow Length	Neighbors Asked	Weights	Total Traffic Input	Total Traffic Output	Total Waiting Time	Average Total Waiting Time	Total Accidents	Priority Vehicles Count	Priority Average Waiting Time
allow-priority?	4	1	-	-	10093	10072	94408	9.35	0	203	1.10
allow-priority?	3	1	-	-	10012	9999	89308	8.92	0	206	1.11
allow-priority?	5	5	-	-	201	118	819558	4077.40	0	2	4918.50
allow-priority?	1	4	-	-	197	109	868870	4410.51	0	2	4899.50



DEMONSTRATION

EditDeleteAdd

Button

slower

ticks: 1

☒ view updates

on ticks

Settings...

grid-size-x 3

grid-size-y 3

setup

go

go once

On Off

auto?

On Off

multi-agent?

On Off

allow-priority?

On Off

adaptive-density?

On Off

adaptive-waiting-time?

On Off

priority-score?

switch all lights

switch current light

Select intersection

speed-limit 5

max-accel 2

max-brake 4

freq-north 100 %

freq-west 100 %

green-length 10

yellow-length 3

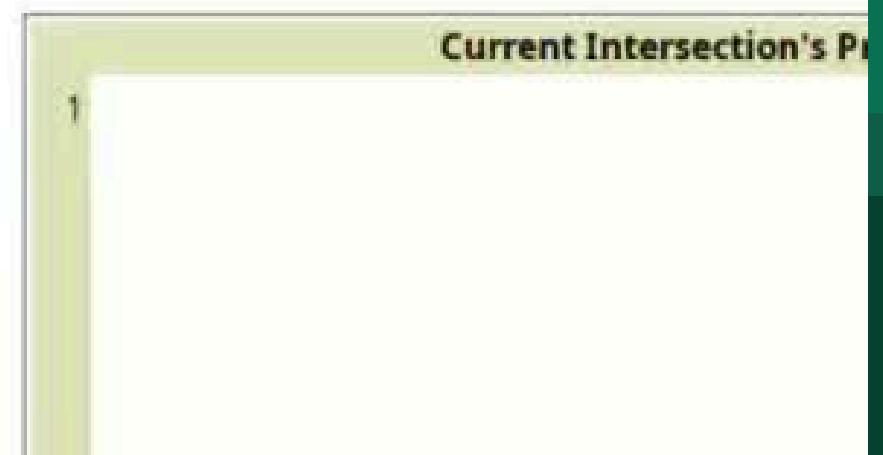
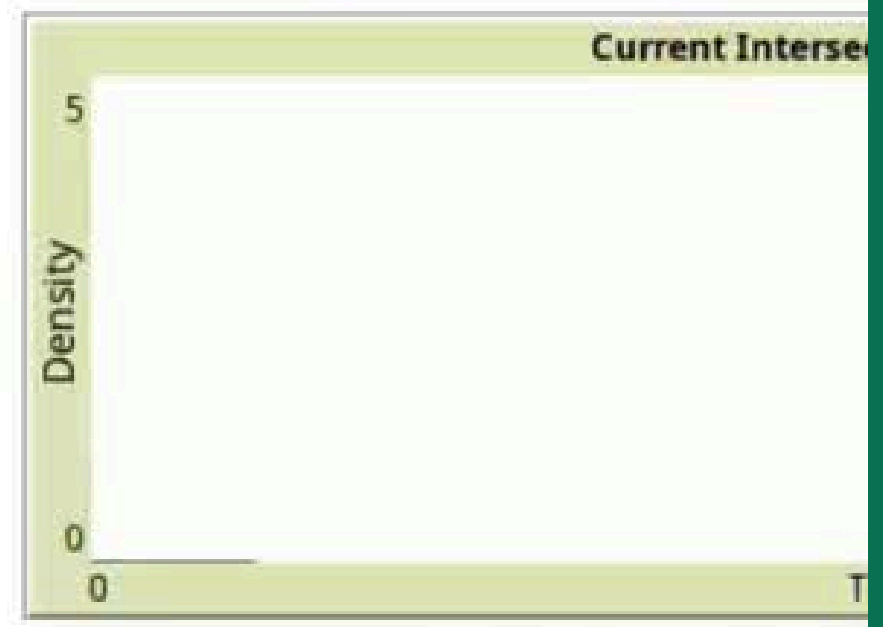
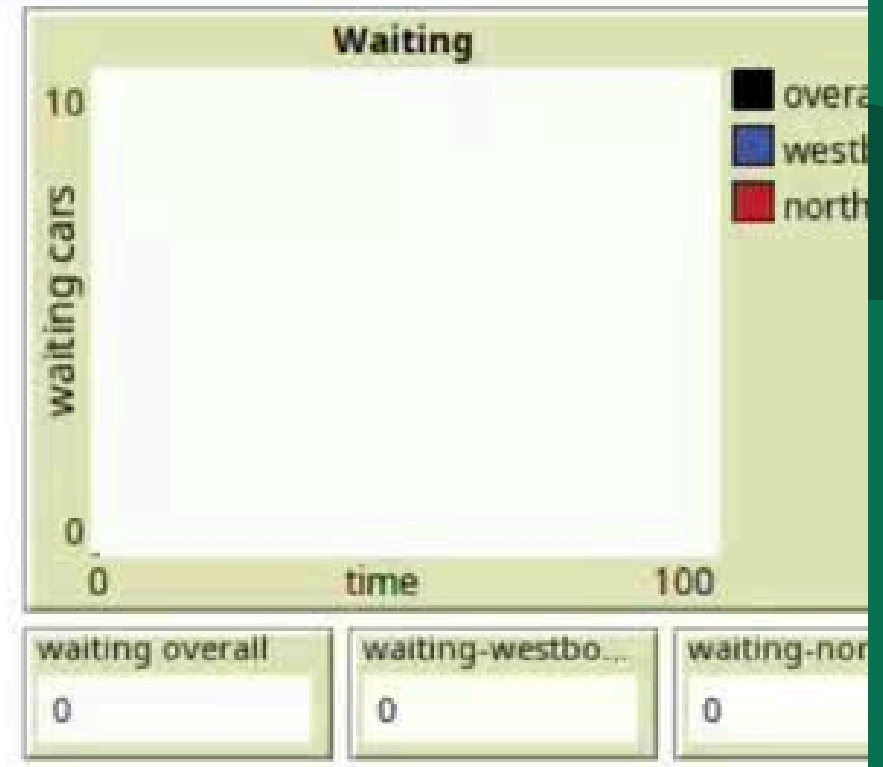
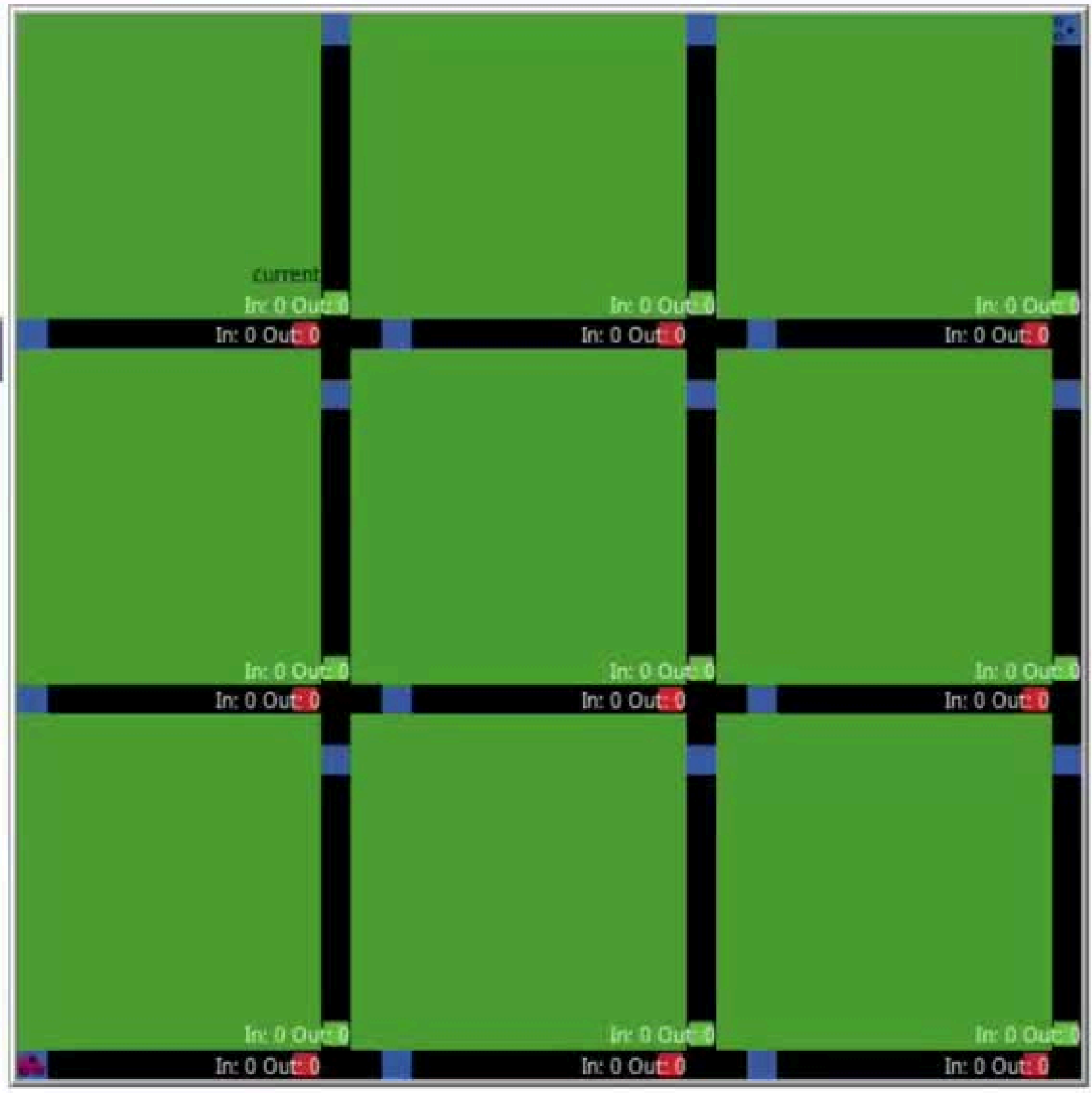
neighbors-asked 2

density-weight 15

waiting-time-weight 15

priority-vehicle-weight 70

lights-time-ticks 50



Total Traffic Input 0

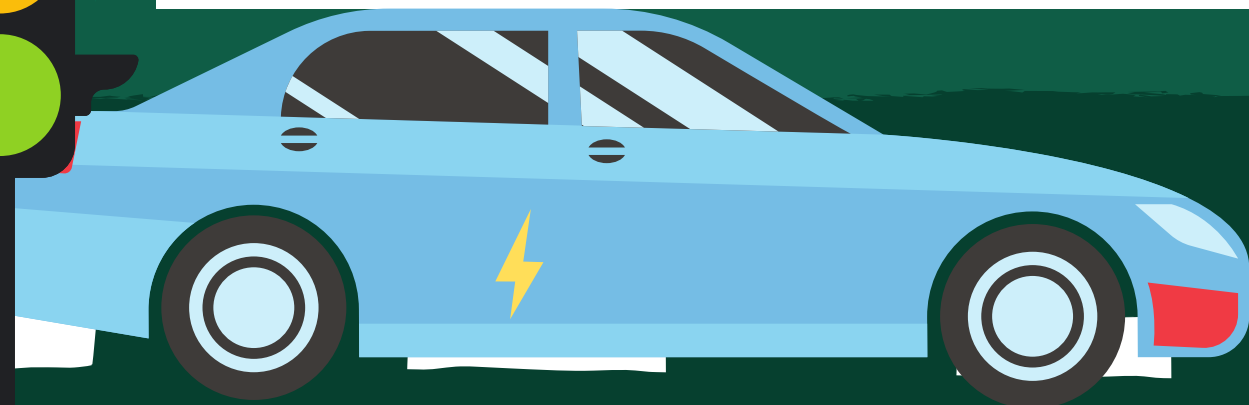
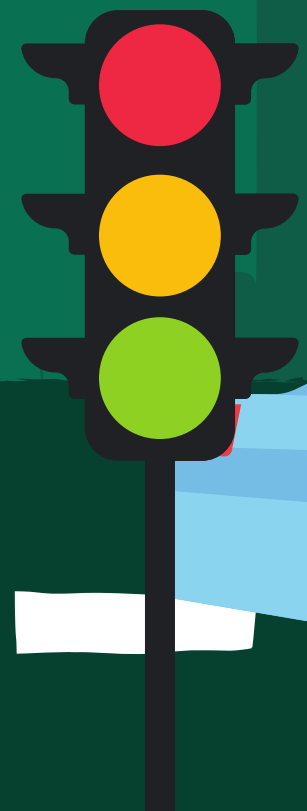
Total Waiting Time 0

Number of accidents 0

CONCLUSIONS



- The results show that the best scenario depends on the metric being prioritized, as different metrics can lead to different outcomes.
- Choosing the most suitable scenario depends on the stakeholders' goals and priorities, with the model helping to guide their decisions.
- As this is a speculative model, it offers flexibility to adapt to various needs and use cases.
- For future work:
 - Explore more advanced algorithms that can deliver more precise results while incorporating broader variables and metrics.
 - Improve the behaviours of the entities



THANK YOU

